

Warm-up 1

What flex activities have you been working on?

What flex activities are you looking forward to working on as we wrap up the course?

Share your thoughts with a neighbor.

Two additional supplementary problems for HW7:

IV. Always, sometimes, or never? Prove if always or never, give an example each way if sometimes.

i) An orthogonal operator on $\mathbb{R}^2$ with determinant 1 is a rotation.

ii) An orthogonal operator on $\mathbb{R}^3$ with determinant 1 is a rotation.

V. Let $A = \begin{pmatrix} x & 1 \\ 0 & x \end{pmatrix}$. Compute each of the following functions of A .

(i) A^2 (ii) A^3 (iii) A^{-1}

Consider the results so far. What do you notice? What do you wonder? Conjecture what will happen for the following functions of A .

(iv) A^{-3} (v) $A^6 - A^4$ (vi) $\sqrt{A}$ (vii) e^A

Check to see if you are correct. What do you conjecture the result of applying an arbitrary function f to A will be? What other interesting cases can you verify? What is going on here?